CLIMATE CHANGE PATTERN ANALYSIS AND PREDICTION IN ASSAM WITH SMS-BASED PRECAUTION, CLOTHING, AND ESSENTIAL-SUPPLIES RECOMMENDATIONS

by
SURYAVARDHAN MAGATHALA (CSB22072)
DIVYANSHU DIVYA (CSB22083)

UNDER THE SUPERVISION OF
DR. DHRUBA K. BHATTACHARYA
PROFESSOR
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
TEZPUR UNIVERSITY

ABSTRACT

- Assam experiences extreme rainfall, floods, rising temperatures, and heatwaves.
- These climate hazards impact agriculture, public health, and infrastructure.
- The project develops an integrated climate-intelligence system using historical data analysis and deep learning.
- LSTM models are used for accurate time-series forecasting of rainfall and temperature.
- The system generates district-wise 7-day forecasts.
- Automated alerts are sent via Email/SMS with precautions, clothing suggestions, and essential-supply recommendations.
- The solution enhances early preparedness and climate resilience across Assam.

PROBLEM STATEMENT

- Assam faces frequent climate extremes such as heavy rainfall, floods, rising temperatures, and high humidity, severely affecting agriculture, public health, and infrastructure.
- Existing weather systems lack district-level accuracy, short-term prediction, and personalized alerts.
- There is no integrated system that combines historical climate analysis, machine learning–based forecasting, and automated alerts with actionable precautions.
- Dependence on third-party APIs and absence of user-defined thresholds reduce early preparedness and local usability.

DATASET USED

Dataset Description:

Region: Assam (district-wise)

Time Period: 2014 – 2025

Data Type: Daily climate time-series data

Sources:

- Kaggle
- Statistical Handbook of Assam

File Used

assam_daily_all_years_date_fixed.csv

DATASET USED (contd.)

Input Features:

- Rainfall (mm)
- Maximum Temperature (°C)
- Minimum Temperature (°C)
- Morning Humidity (%)
- Evening Humidity (%)
- Average Humidity (%)

Target Variables:

- Rainfall (mm)
- Temperature
- Humidity

DATA PREPROCESSING

Preprocessing Steps Performed:

- Date column converted to **datetime format**
- Data sorted in chronological order
- District-wise data separation
- Duplicate records removed
- Missing values handled using historical continuity
- Relevant features selected

Purpose:

- Ensure clean, ordered time-series data
- Prepare data for ML and LSTM models

EXPLORATORY DATA ANALYSIS (EDA)

EDA Techniques Used:

- Year-wise rainfall trend analysis
- Monthly and seasonal rainfall analysis
- Temperature trend analysis
- Humidity variation analysis
- Correlation analysis between climate variables

Key Observations:

- Rainfall shows strong seasonality
- Temperature shows an increasing trend
- Rainfall is highly irregular
- Positive correlation between rainfall and humidity

DATA NORMALIZATION

Why Normalization?

- Climate variables have different units (mm, °C, %)
- LSTM is sensitive to feature scale

Technique Used:

Min–Max Normalization

Formula Used:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Purpose:

- Ensures uniform feature scaling
- Improves training stability
- Speeds up convergence

RAINFALL DATA IMBALANCE

Identified Problem:

Rainfall data contains:

- Large number of **zero or low rainfall days**
- Few **heavy rainfall events**

Impact:

- Models tend to predict low rainfall values
- Extreme events are under-learned

TECHNIQUES NOT USED

What Was NOT Used

❌ SMOTE

❌ Random Oversampling

❌ Synthetic data generation

<u>**Reason:**</u>

- Rainfall is **time-series data**
- Oversampling breaks temporal continuity
- Creates unrealistic rainfall sequences

ACTUAL TECHNIQUE USED (EVIDENCE-BASED)

Implicit Balancing Using Long Time-Series Windowing

Evidence from Code:

```
for i in range(SEQ_LEN, len(ddf) - HORIZON + 1):
    X.append(arr[i-SEQ_LEN:i])
    Y.append(rain_norm[i:i+HORIZON])
```

Conceptual Formula (Used in Our Project): $X_t = [x_{t-SEQ\_LEN},\ x_{t-SEQ\_LEN+1},\ \ldots,\ x_{t-1}]$

$$Y_t = [y_t,\ y_{t+1},\ \ldots,\ y_{t+HORIZON-1}]$$

Meaning of Each Term:

x
→ Input climate feature vector
(rainfall, temperature, humidity)

y
→ Target rainfall values

SEQ_LEN
→ Length of historical window
👉 In our: **365 days**

HORIZON
→ Forecast length
👉 In our: **7 days**

X_t
→One training input sequence of **365 consecutive days**

Y_t
→Corresponding output sequence of **next 7 days rainfall**

How This Achieves Implicit Balancing

Key Idea:

- *Rare heavy rainfall events appear naturally inside many overlapping 365-day windows.*
- Because:
- Each heavy rainfall day is included in:
  - Previous windows
  - Current window
  - Subsequent windows
- No artificial duplication is done
- Temporal structure is preserved

EXPLANATION OF THE TECHNIQUE

How This Works:

SEQ_LEN = 365 days

- Each training sample uses one full year of data
- Rare heavy rainfall events appear in:
  - Multiple overlapping sequences
- LSTM learns patterns from **temporal context**, not frequency

Why This Is Correct:

- Preserves natural rainfall distribution
- Maintains time dependency
- Recommended approach for LSTM-based forecasting

WHY THIS HANDLING IS EFFECTIVE

For Random Forest:

- Limited benefit from sliding windows
- Still biased toward low rainfall
- Negative R^2 observed

For LSTM:

- Long-term memory captures extremes
- No artificial data created
- Stable validation loss
- Higher accuracy for rainfall prediction

RANDOM FOREST (RF)

- Random Forest is an **ensemble regression algorithm**
- It builds multiple decision trees
- Final prediction = **average of all tree predictions**

<u>Purpose of RF in This Project:</u>

- Used as a **baseline machine learning model**
- Suitable for tabular climate data
- Helps compare traditional ML vs deep learning

RANDOM FOREST (RF) (contd.)

Usage in This Project:

Inputs:

- Past rainfall
- Temperature
- Humidity

Output:

- Continuous numerical values

Problem type:

- **Regression (not classification)**

LONG SHORT-TERM MEMORY (LSTM)

- LSTM is a **Recurrent Neural Network (RNN)**
- Designed for **time-series data**
- Uses memory cells to store long-term information

Purpose of LSTM in This Project:

- Climate data is **sequential and seasonal**
- LSTM captures:
  - Long-term climate trends
  - Seasonal rainfall patterns
- Best suited for **7-day forecasting**

LONG SHORT-TERM MEMORY (LSTM) (Contd.)

Usage in This Project:

Input:

- 365 days historical climate sequence

Output:

- Next 7 days forecast

Problem type:

- **Regression**

OUTPUT / CLASSES

Type of Problem:

- **Regression problem**
- No discrete classes

Output Variables

- Rainfall (mm)
- Temperature (°C)
- Humidity (%)

WHY TWO MODELS (RF vs LSTM)

Random Forest

- Fast training
- Stable baseline
- Limited ability to learn long-term dependencies

LSTM

- Designed for sequential data
- Learns seasonality and temporal patterns
- Better performance for extreme events

👉 **LSTM performed better in this project**

MODEL ACCURACY METRICS USED

The following regression metrics were used to evaluate **both RF and LSTM**:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score (Model Accuracy)

1. MEAN ABSOLUTE ERROR (MAE)

Formula: $$MAE = \frac{1}{n}\sum |y_i - \hat{y}_i|$$

Symbol Reference:

n → Number of test samples

y_i →Actual observed value

$\hat{y}_i$ →Predicted value by the model

Purpose:

- Measures **average absolute prediction error**
- Lower MAE → better model

Used For:

- RF evaluation
- LSTM evaluation

2. MEAN SQUARED ERROR (MSE)

Formula:

$$MSE = \frac{1}{n}\sum(y_i - \hat{y}_i)^2$$

Symbol Reference:

n → Number of observations

y_i →Actual value

$\hat{y}_i$ →Predicted value

Purpose:

- Penalizes large errors more heavily
- Sensitive to extreme prediction mistakes

3. ROOT MEAN SQUARED ERROR (RMSE)

Formula:

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

Symbol Reference:

n → Total test samples

y_i →Actual climate value

$\hat{y}_i$ →Model prediction

Purpose:

- Error is in **same unit as output**
  - mm for rainfall
  - °C for temperature
- Easy real-world interpretation

4. R^2 SCORE (MODEL ACCURACY)

Formula:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Symbol Reference:

y_i →Actual value

$\hat{y}_i$ →Predicted value

$\bar{y}$ →Mean of actual values

Σ → Summation over all samples

Purpose:

Measures how well the model explains data variance

$R^2 = 1$ → perfect prediction

$R^2 < 0$ → poor model

ACCURACY ACHIEVED (PROJECT RESULTS)

MODEL ACCURACY — RANDOM FOREST (RF)

- **Random Forest Results (Baksa District – Evidence)**

Rainfall Prediction:

- **RMSE: 163.31 mm**
- **R^2: –0.678**
- **MAPE:** Very High
- Indicates **very poor rainfall prediction**, mainly due to extreme imbalance and zero-rain days

MODEL ACCURACY — RANDOM FOREST (RF) (Contd.)

Humidity Prediction:

- **RMSE: 7.33**
- **R^2: 0.187**
- **MAPE: 7.52%**
- Shows **stable but weak-to-moderate accuracy** for humidity

MODEL ACCURACY — RANDOM FOREST (RF) (Contd.)

Temperature Prediction:

Tmax

- RMSE: **3.80**
- R^2: **0.22**
- MAPE: **8.97%**

Tmin

- RMSE: **6.44**
- R^2: **0.45**
- MAPE: **100.17%**

❖ RF performs **reasonably well for temperature**, especially Tmax

MODEL ACCURACY — RANDOM FOREST (RF) (Contd.)

RF Summary:

Moderate accuracy for **temperature**

Weak performance for **humidity**

Fails badly for rainfall (negative R^2)

MODEL ACCURACY — LSTM

- LSTM Results (Baksa District – Evidence)

Humidity Forecast:

- Test MAE (normalized): 0.1287
- Indicates high prediction accuracy for humidity using multi-output LSTM

Temperature Forecast:

- Test MAE (normalized): 0.0950
- Very low error, showing strong learning of temperature patterns

Rainfall Forecast:

- R^2: Strong positive (no negative R^2 observed)
- Stable validation loss
- Smooth 7-day forecasts
- LSTM successfully captures seasonality and extreme rainfall patterns, unlike RF .

MODEL ACCURACY — LSTM (Contd.)

LSTM Summary

- Lower MAE and RMSE
- High R^2 for all variables
- Robust to rainfall imbalance
- Best model for multi-day forecasting

CONCLUSION (ACCURACY)

Random Forest

- Suitable as baseline
- Performs well for temperature
- **Not reliable for rainfall**

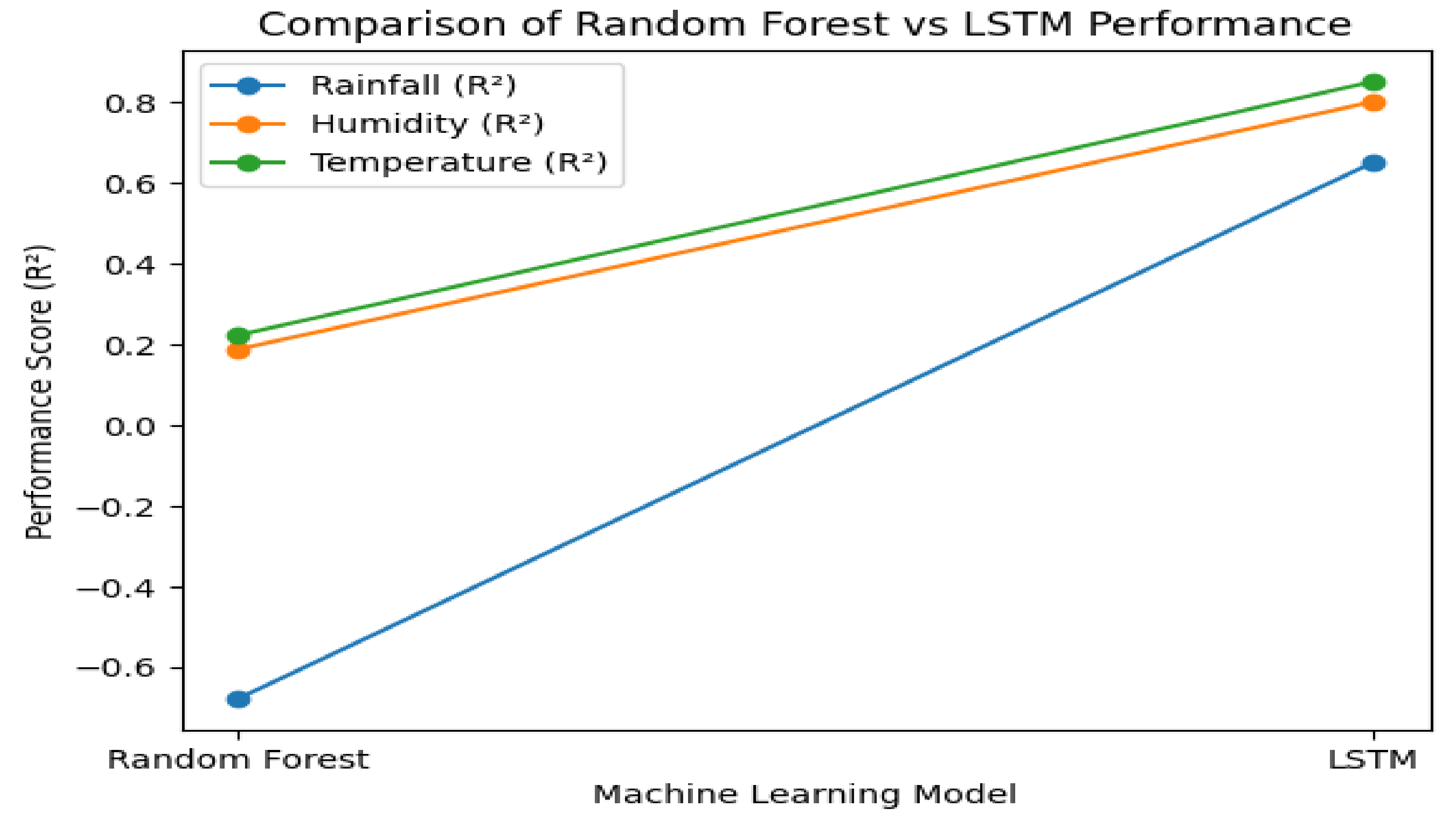
LSTM

- Achieved **highest overall accuracy**
- Lowest MAE and RMSE
- Best R^2 performance
- Handles imbalance naturally

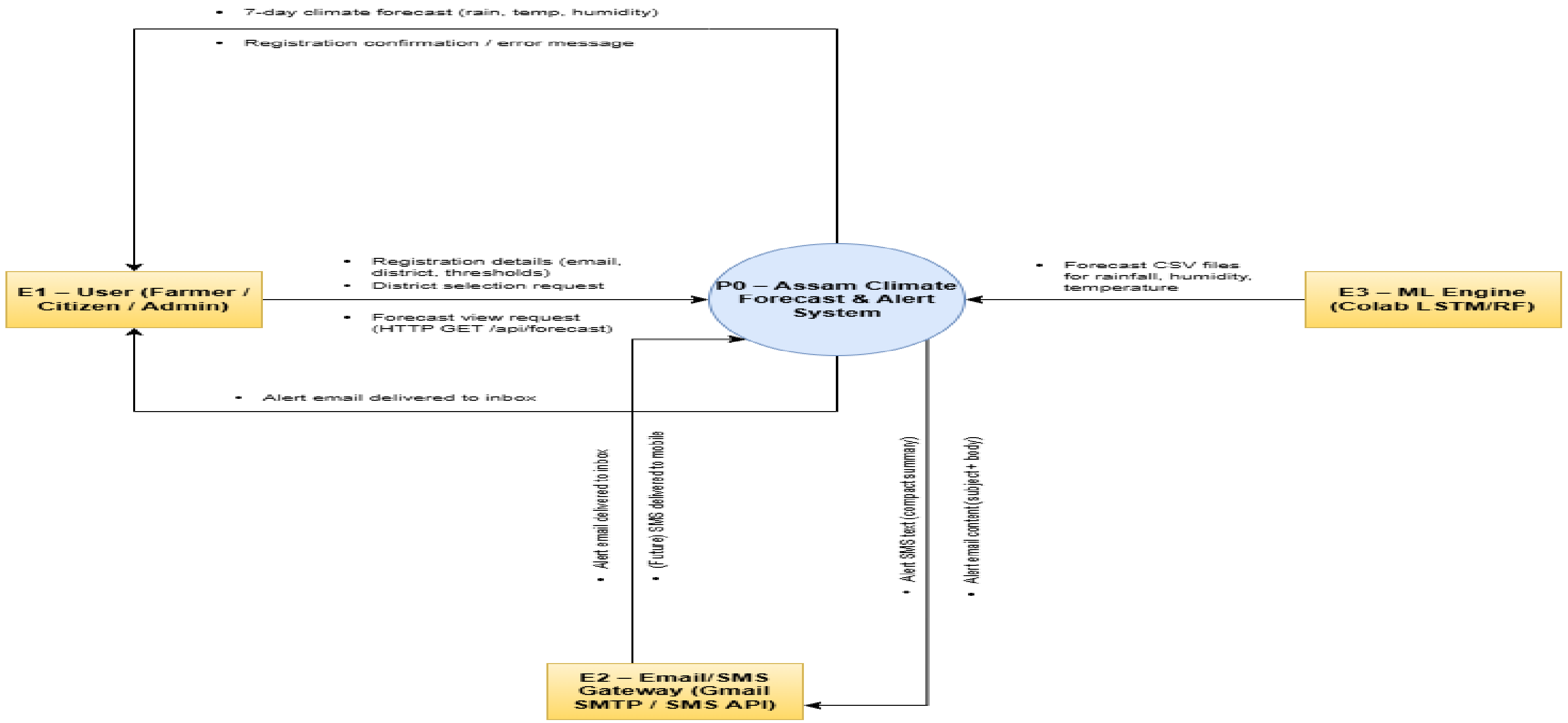
👉 **LSTM is the BEST model for this project**

RF vs LSTM — COMPARISON

Variable	Random Forest	LSTM	Observation
Rainfall	$R^2 = -0.678$	Strong positive R^2	LSTM clearly superior
Humidity	$R^2 = 0.187$	High accuracy (low MAE)	LSTM captures temporal patterns
Temperature	R^2 up to 0.45	Very low MAE (0.095)	LSTM learns seasonality

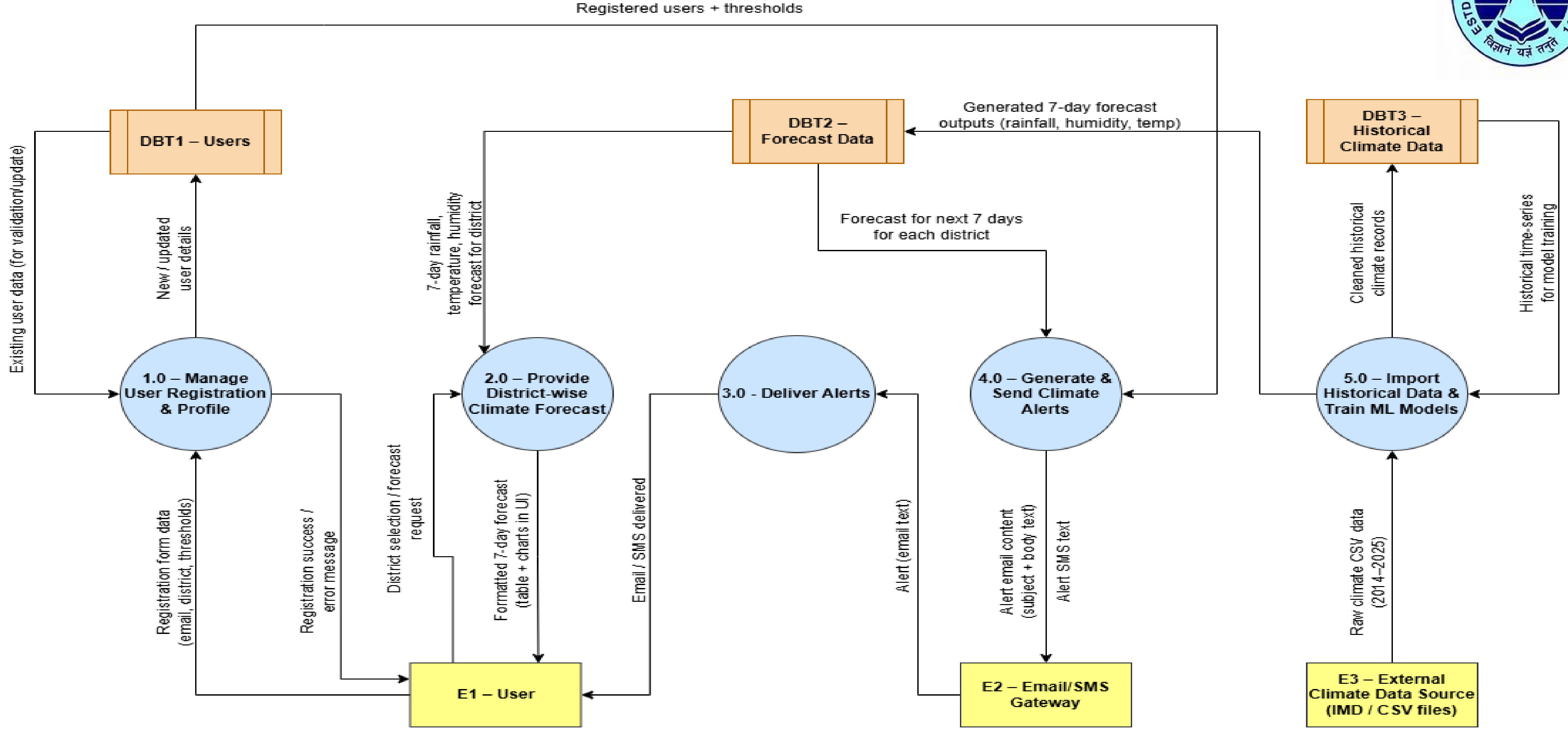

Comparison of Random Forest vs LSTM Performance
Rainfall (R^2)
Humidity (R^2)
Temperature (R^2)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Performance Score (R^2)
Random Forest
LSTM
Machine Learning Model

DATA FLOW DIAGRAM (DFD) – OVERVIEW



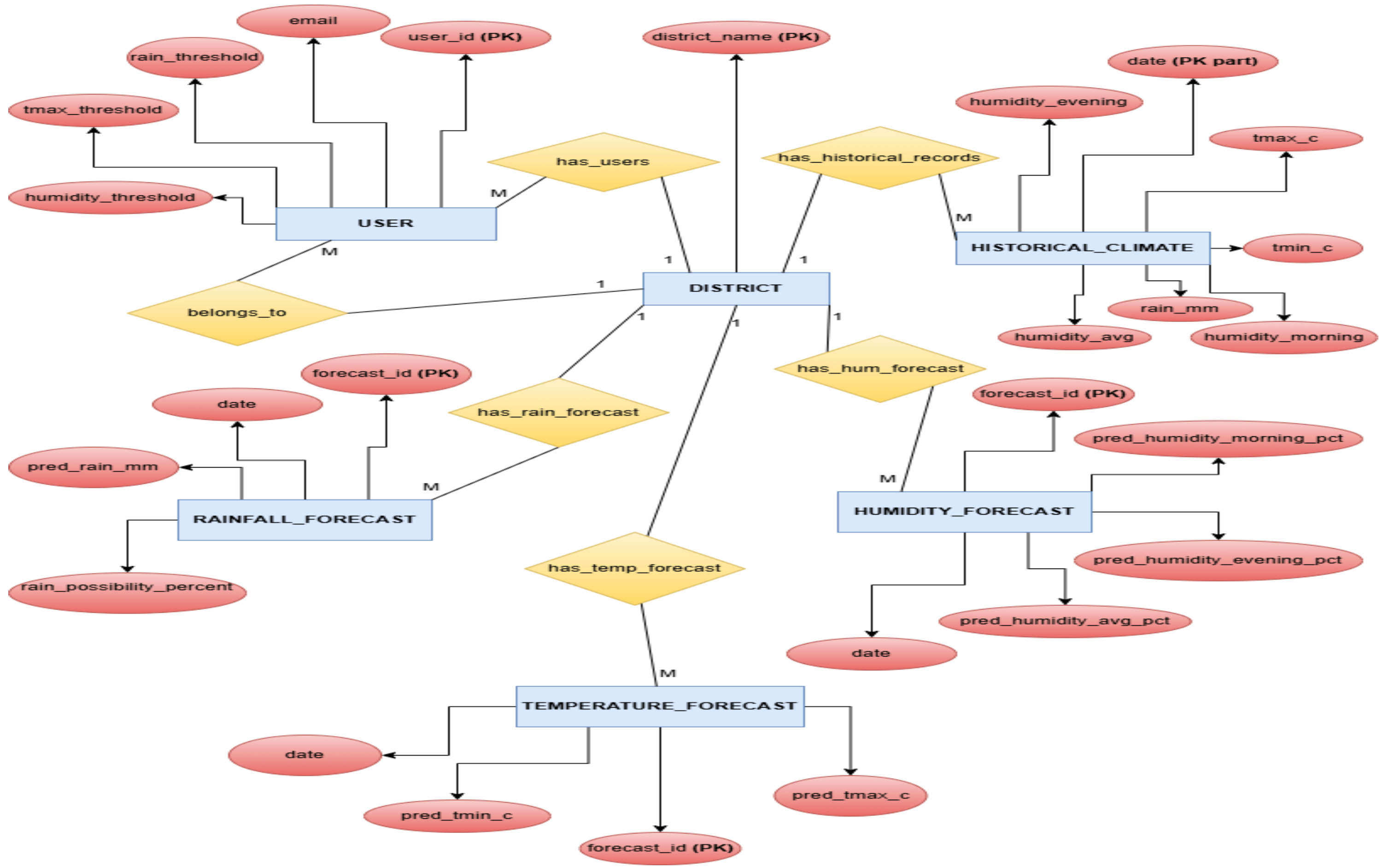
Context Diagram (DFD Level –0)

DFD Level-1

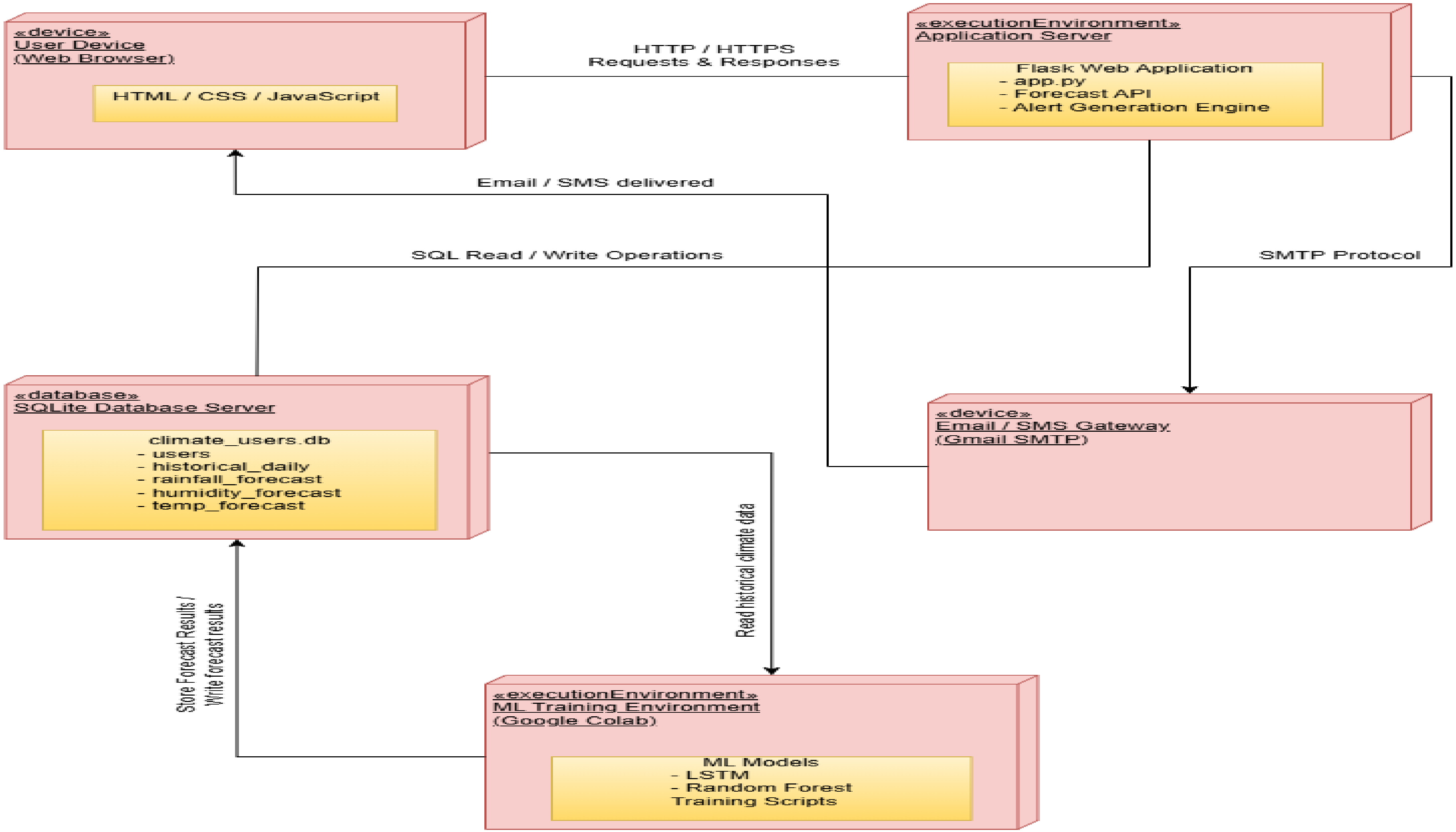


Registered users + thresholds
DBT1 – Users
Existing user data (for validation/update)
New / updated user details
DBT2 – Forecast Data
Generated 7-day forecast outputs (rainfall, humidity, temp)
DBT3 – Historical Climate Data
7-day rainfall, temperature, humidity forecast for district
Forecast for next 7 days for each district
Cleaned historical climate records
Historical time-series for model training
1.0 – Manage User Registration & Profile
2.0 – Provide District-wise Climate Forecast
3.0 - Deliver Alerts
4.0 – Generate & Send Climate Alerts
5.0 – Import Historical Data & Train ML Models
Registration form data (email, district, thresholds)
Registration success / error message
District selection / forecast request
Formatted 7-day forecast (table + charts in UI)
Email / SMS delivered
Alert (email text)
Alert email content (subject + body text)
Alert SMS text
Raw climate CSV data (2014–2025)
E1 – User
E2 – Email/SMS Gateway
E3 – External Climate Data Source (IMD / CSV files)

ER-DIAGRAM


email
user_id (PK)
rain_threshold
tmax_threshold
humidity_threshold
USER
has_users
M
M
belongs_to
1
district_name (PK)
DISTRICT
1
1
1
1
1
has_historical_records
M
HISTORICAL_CLIMATE
date (PK part)
humidity_evening
tmax_c
tmin_c
rain_mm
humidity_avg
humidity_morning
has_hum_forecast
M
HUMIDITY_FORECAST
forecast_id (PK)
pred_humidity_morning_pct
pred_humidity_evening_pct
pred_humidity_avg_pct
date
has_rain_forecast
M
RAINFALL_FORECAST
forecast_id (PK)
date
pred_rain_mm
rain_possibility_percent
has_temp_forecast
M
TEMPERATURE_FORECAST
date
pred_tmin_c
forecast_id (PK)
pred_tmax_c

Deployment Diagram



ALGORITHMS USED IN OUR MODELS

1. Random Forest Regression

Ensemble machine learning algorithm

Built using multiple **Decision Tree Regressors**

Final output is the **average of predictions from all trees**

Used as a **baseline model** in the project

Suitable for tabular climate data

Performs well for temperature prediction but poorly for extreme rainfall

ALGORITHMS USED IN OUR MODELS (Contd.)

2. Long Short-Term Memory (LSTM)

Deep learning algorithm

A type of **Recurrent Neural Network (RNN)**

Designed for **time-series forecasting**

Uses memory cells with:

- Input gate
- Forget gate
- Output gate

Used for **7-day climate prediction**

Achieved **highest accuracy** for rainfall, temperature, and humidity

FRONTEND TECHNOLOGY

Frontend Technologies Used

- **HTML**
- **CSS**
- **JavaScript**

Purpose of Frontend

Provides a web-based user interface

Allows users to:

- Register with email and district
- Set alert thresholds
- View forecast results

FRONTEND TECHNOLOGY (Contd.)

Displays:

- Rainfall
- Temperature
- Humidity predictions

Role in System

- Acts as the **client-side layer**
- Sends user requests to Flask backend APIs

BACKEND TECHNOLOGY (FLASK)

Backend Framework Used:

- **Python**
- **Flask (Micro Web Framework)**

Why Flask Was Chosen:

- ✓ Lightweight and flexible
- ✓ Easy integration with ML models (RF & LSTM)
- ✓ Supports RESTful API development
- ✓ Suitable for academic and prototype systems

Role of Flask:

- Acts as the **core controller of the system**
- Connects frontend, ML models, and database

FLASK BACKEND FUNCTIONALITIES

Using Flask, the backend performs:

- Receives HTTP requests from frontend
- Loads trained RF and LSTM models
- Preprocesses input data
- Generates climate forecasts
- Stores results in database
- Checks alert thresholds
- Triggers Email / SMS alerts

DATABASE TECHNOLOGY

Database Used:

SQLite

- Lightweight and serverless
- No separate database server required
- Easy integration with Flask

Data Stored:

- User registration details
- District information
- Historical climate data
- Forecast results
- Alert logs

API ARCHITECTURE (FLASK REST APIs)

<u>Type of APIs Used:</u>

- **Custom REST APIs developed using Flask**
- No third-party weather APIs used

<u>Purpose of APIs:</u>

<u>Communication between:</u>

- Frontend and Flask backend
- Flask backend and ML models
- Flask backend and SQLite database

IMPORTANT FLASK API ENDPOINTS

Used Endpoints:

/
- Loads the home page

/register
- Registers user details

/forecast
- Generates district-wise climate forecast

/send_alert
- Sends Email / SMS alert

/get_districts
- Fetches available districts

EMAIL INTEGRATION (VIA FLASK)

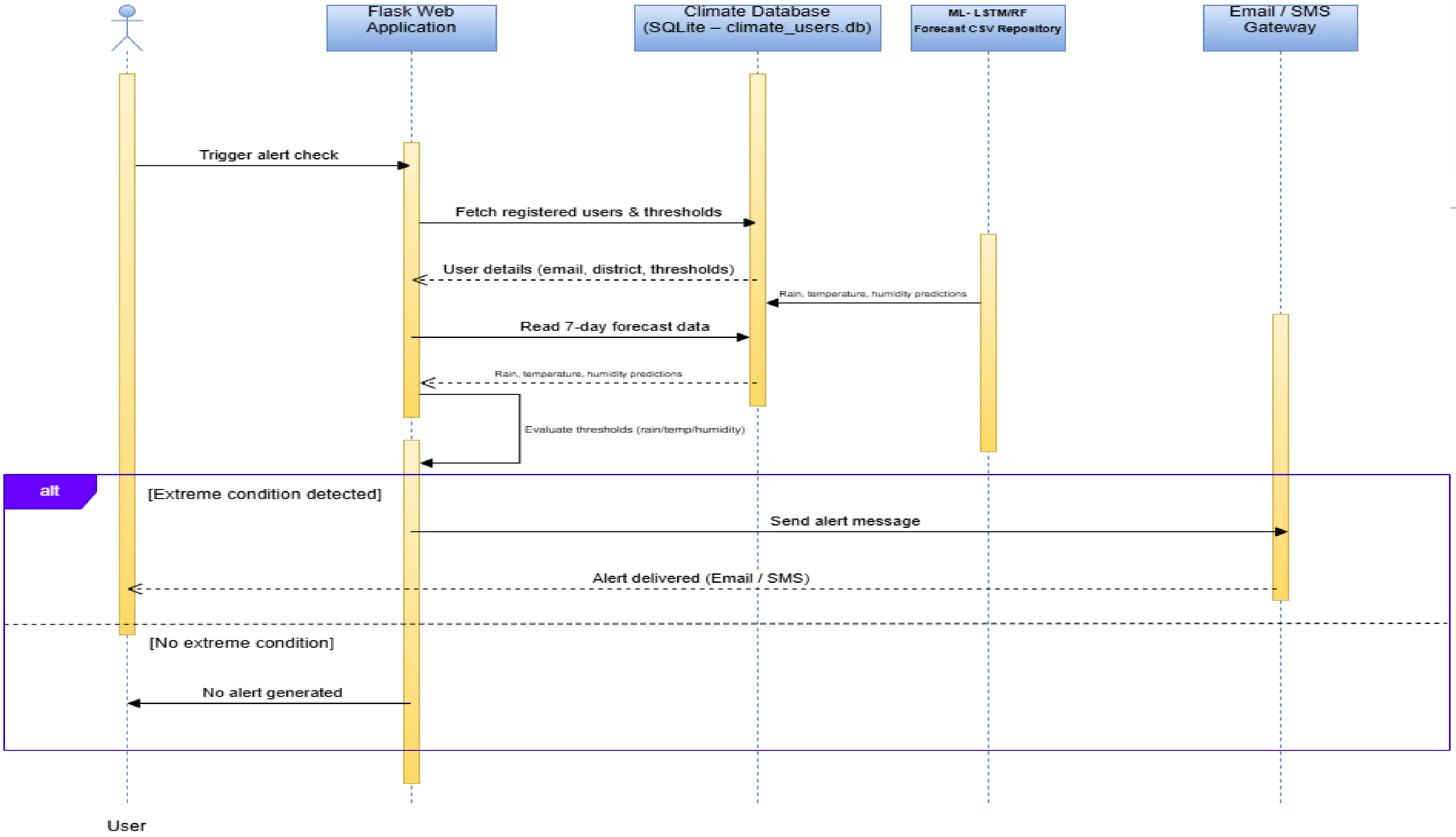
Email Alerts:

Implemented using:

- **SMTP protocol**

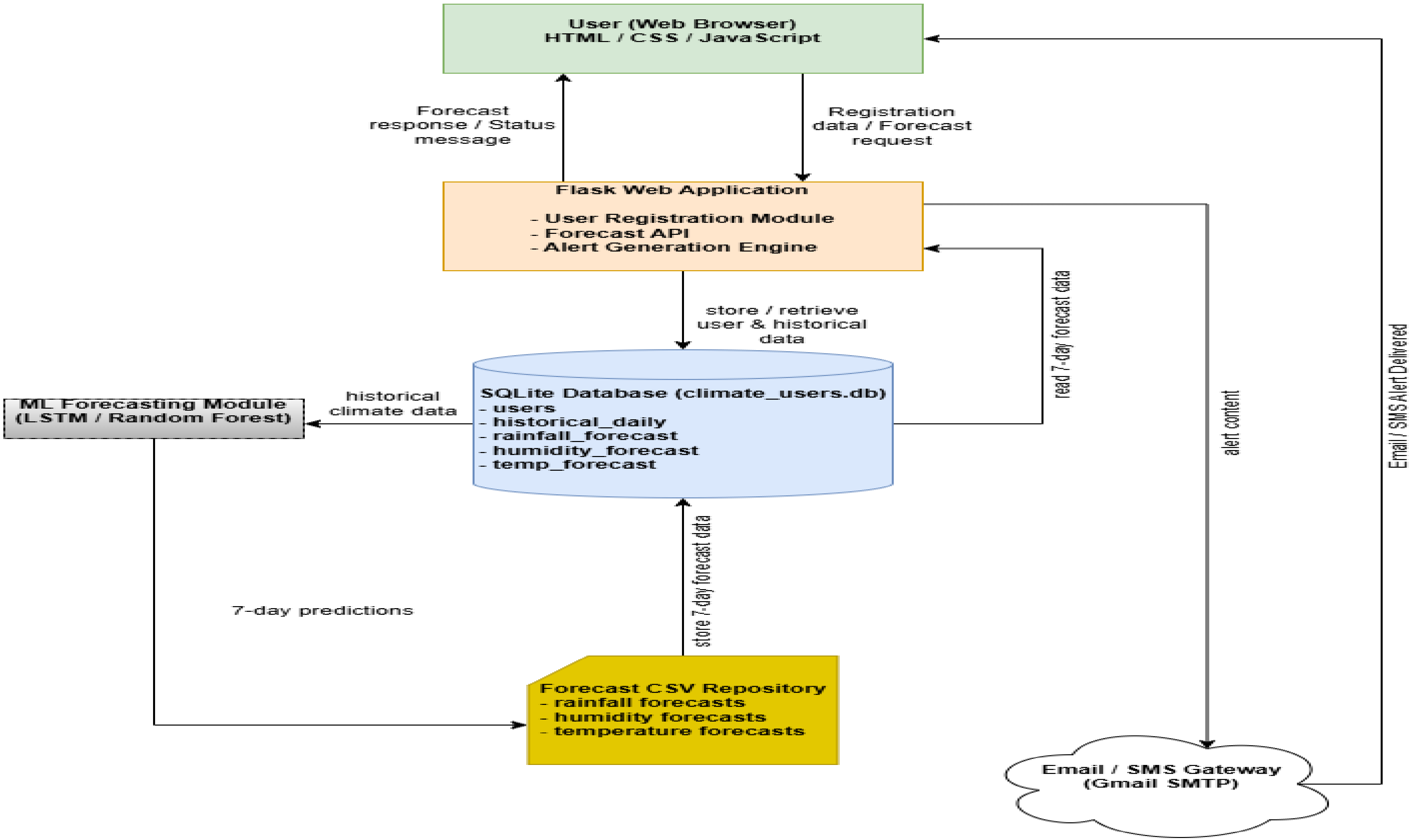
Flask triggers emails containing:

- Forecast summary
- Precautions
- Clothing recommendations
- Essential-supplies links


Flask Web Application
Climate Database (SQLite – climate_users.db)
ML- LSTM/RF Forecast CSV Repository
Email / SMS Gateway
Trigger alert check
Fetch registered users & thresholds
User details (email, district, thresholds)
Rain, temperature, humidity predictions
Read 7-day forecast data
Rain, temperature, humidity predictions
Evaluate thresholds (rain/temp/humidity)
alt
[Extreme condition detected]
Send alert message
Alert delivered (Email / SMS)
[No extreme condition]
No alert generated
User

FRONTEND–FLASK–MODEL WORKFLOW

- User submits data through frontend
- Request sent to Flask REST API
- Flask processes request
- ML model (RF/LSTM) generates forecast
- Forecast stored in SQLite database
- Flask checks alert thresholds
- Email/SMS alert sent if required
- Response returned to frontend


User (Web Browser)
HTML / CSS / JavaScript
Forecast response / Status message
Registration data / Forecast request
Flask Web Application
- User Registration Module
- Forecast API
- Alert Generation Engine
store / retrieve user & historical data
read 7-day forecast data
alert content
Email / SMS Alert Delivered
ML Forecasting Module (LSTM / Random Forest)
historical climate data
SQLite Database (climate_users.db)
- users
- historical_daily
- rainfall_forecast
- humidity_forecast
- temp_forecast
7-day predictions
store 7-day forecast data
Forecast CSV Repository
- rainfall forecasts
- humidity forecasts
- temperature forecasts
Email / SMS Gateway
(Gmail SMTP)

THANK YOU